



Proposal Document

Master of Science (M.Sc)

Machine Learning

Major: Data Science (DS)

Topic: Chest X-Ray Pneumonia Detection System

Author: Ivan Logutov

Matrikel-Number: 80086466

Supervisor: Raja Hashim Ali

Submitted on: 03.06.2026

1. Project Title

Field name: Medical Image Analysis

Project idea title: Chest X-Ray Pneumonia Detection System

2. Background and Motivation

Pneumonia is one of the leading causes of death among children under five worldwide, and chest radiography (X-ray) is the most common and accessible tool for its diagnosis. However, reading chest X-rays requires trained radiologists, who are scarce in many regions, and even experts show inter-observer variability. Delays or errors in diagnosis can be fatal, because untreated bacterial pneumonia progresses quickly.

Convolutional Neural Networks (CNNs) have achieved radiologist-level performance on several medical-imaging tasks. An automated screening tool that flags likely pneumonia cases could act as a decision-support system - prioritising urgent cases and reducing the chance that a sick patient is overlooked.

This project applies the supervised-learning, neural-network and CNN concepts from the course to a real, clinically motivated image-classification problem, and studies whether transfer learning can compensate for the limited size of medical datasets.

3. Problem Statement

Given a single grayscale chest X-ray image, automatically classify it as NORMAL or PNEUMONIA.

Formally, this is a supervised binary image-classification task:

Learn a function $f(\text{image}) \rightarrow \{0 = \text{NORMAL}, 1 = \text{PNEUMONIA}\}$ that maximises diagnostic sensitivity (recall) while keeping false alarms acceptable.

Because a False Negative (a pneumonia patient classified as healthy) is clinically far more dangerous than a False Positive (a healthy patient flagged for review), recall/sensitivity is treated as the primary metric, with F1 as the secondary metric.

4. Selected Dataset and Dataset Link

Dataset: Chest X-Ray Images (Pneumonia) - Kermany et al., Guangzhou Women and Children's Medical Center

Link: <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>

Kaggle slug: paultimothymooney/chest-xray-pneumonia

5. Dataset Description

The dataset contains anterior-posterior chest X-ray images of pediatric patients (ages 1-5). Images were quality-screened and graded by expert physicians. It is organised into train/, val/, and test/ folders, each containing NORMAL and PNEUMONIA subfolders.

Number of images: 5,856 total (train: 5,216 - original val: 16 - test: 624)

Number of classes: 2 - NORMAL (0) and PNEUMONIA (1). The PNEUMONIA class mixes bacterial and viral cases (not separated in this binary task)

Image type: Grayscale chest X-ray photographs in JPEG format, variable resolution (sampled mean approx. 1440 x 1099 px), single channel

Class distribution:

Split	NORMAL	PNEUMONIA	Total	Ratio (N : P)
Train (original)	1,341	3,875	5,216	approx. 1 : 2.9
Test	234	390	624	approx. 1 : 1.7
Original val	8	8	16	1 : 1

Two data issues are explicitly addressed in the methodology: (i) the class imbalance in the training set, and (ii) the unusable original validation split (only 16 images - rebuilt via stratified 80/20 split).

6. Research Questions

1. RQ1. Can a CNN reliably detect pneumonia from chest X-rays, achieving high sensitivity (recall ≥ 0.95) on an unseen test set?
 2. RQ2. Does transfer learning (VGG16) outperform a CNN trained from scratch when the training data is limited (~5k images), and on which metrics?
 3. RQ3. How does handling class imbalance (class weights) affect the model's recall and the balance between false negatives and false positives?
 4. RQ4. Do the models base their decisions on clinically relevant regions (the lung fields), as assessed with Grad-CAM explanations?
-

7. Expected Results

- Both models are expected to learn the task well, with the transfer-learning model (VGG16) expected to outperform the custom CNN, especially on AUC, precision and F1, because its ImageNet-pretrained features generalise better on a small dataset.
- Target performance on the held-out test set: recall ≥ 0.95 , AUC ≥ 0.95 , accuracy in the 0.85-0.92 range for the best model.
- Applying class weights is expected to push the models toward high sensitivity (few false negatives), at the cost of some additional false positives - the desired trade-off for a clinical screening tool.
- Grad-CAM heatmaps are expected to concentrate on the lung fields for correct predictions, providing qualitative evidence that the model is clinically plausible.

Preliminary results (obtained during development):

Model	Accuracy	Precision	Recall	F1	AUC
Custom CNN	0.728	0.697	0.997	0.821	0.930
VGG16 (Transfer Learning)	0.896	0.868	0.982	0.922	0.970

8. Proposed Methodology

8.1 Data Preparation

- Resizing & normalisation: all images resized to 224x224; grayscale replicated to 3 channels (RGB) to match the transfer-learning backbone; pixels normalised to [0, 1] via $\text{rescale} = 1/255$.
- Train/validation/test split: the broken original val/ is discarded; a stratified 80/20 split of the training data rebuilds a meaningful validation set; the original test/ is kept untouched. Because Kaggle input is read-only, files are not copied - paths and labels are collected into DataFrames and fed via `flow_from_dataframe`.
- Data augmentation (training only, conservative for medical imaging): horizontal flip, +/-10 deg rotation, <=10% width/height shift and zoom. No vertical flip, shear, or heavy colour jitter - anatomy must stay upright. Validation/test use rescaling only.
- Class imbalance: handled with `class_weight = "balanced"`, penalising errors on the minority NORMAL class more heavily.

8.2 Training Configuration

- Optimizer: Adam | Loss: Binary Cross-Entropy | Batch size: 32 | Epochs: up to 30
- Callbacks: EarlyStopping and ModelCheckpoint monitoring `val_auc` (a threshold-free metric robust to the trivial "always-PNEUMONIA" solution that maximises recall), plus `ReduceLROnPlateau` on `val_loss`. Best weights are reloaded before evaluation.
- Reproducibility: all random seeds fixed to 42 (Python / NumPy / TensorFlow).

9. CNN Model Design (From Scratch)

A compact 4-block convolutional network (~0.5 M parameters):

```
Input (224x224x3)
-> [Conv2D(32, 3x3) -> BatchNorm -> ReLU -> MaxPool]
-> [Conv2D(64, 3x3) -> BatchNorm -> ReLU -> MaxPool]
-> [Conv2D(128, 3x3) -> BatchNorm -> ReLU -> MaxPool]
-> [Conv2D(256, 3x3) -> BatchNorm -> ReLU -> MaxPool]
-> GlobalAveragePooling2D
```

```
-> Dense(512, ReLU) -> Dropout(0.5)
-> Dense(1, Sigmoid)
```

Filters double each block (32->64->128->256), progressively building richer feature maps.

GlobalAveragePooling (instead of a large Flatten) and Dropout(0.5) regularise the model and reduce overfitting on the limited dataset.

10. Transfer Learning Model (VGG16)

Backbone: VGG16 pretrained on ImageNet, include_top=False, input 224x224x3.

Custom head: GlobalAveragePooling -> Dense(256, ReLU) -> Dropout(0.4) -> Dense(1, Sigmoid).

Two-phase training:

1. Feature extraction - base frozen (10 epochs, lr = 1e-4): only the custom head is trained, allowing the network to adapt the task without disturbing ImageNet weights.
2. Fine-tuning - last 4 convolutional layers unfrozen (10 epochs, lr = 1e-5): the highest-level features are gently adapted to the medical domain at a 10x smaller learning rate, avoiding catastrophic forgetting.

VGG16 is chosen because it has a well-understood architecture, is available in Keras, and its modest depth allows fine-tuning even on limited compute. Its ImageNet-pretrained low-level edge and texture detectors are expected to transfer well to grayscale X-ray images replicated to 3 channels.

11. Evaluation Metrics

All evaluation is performed on the untouched test/ split (624 images, never seen during training or validation):

- Recall / Sensitivity (primary) - minimise missed pneumonia (false negatives). A FN means sending a sick patient home without treatment.
- F1-score (secondary) - harmonic mean of precision and recall; balances the sensitivity-specificity trade-off.
- Precision - fraction of pneumonia predictions that are correct.

- Accuracy - overall correct classifications (reported but secondary due to class imbalance).
- AUC (ROC) - threshold-free discrimination quality; also used as the monitoring metric during training to avoid trivial solutions.
- Confusion matrix - visual breakdown of TP, TN, FP, FN for each model.
- Classification report - per-class precision, recall, F1 and support.
- Grad-CAM (qualitative) - visual explanation of which image regions drive predictions; used to verify clinical plausibility.

12. Expected Figures and Tables

Artifact	Purpose
Class-distribution bar chart	Show the training imbalance between NORMAL and PNEUMONIA classes
Sample-images grid	Visual feel for the appearance of NORMAL vs PNEUMONIA X-rays
Training curves (accuracy & loss) - Custom CNN	Diagnose learning speed, overfitting, and convergence
Training curves (accuracy & loss) - VGG16	Compare learning dynamics with the transfer-learning model
Confusion matrix - Custom CNN	Analyse false-negative vs false-positive trade-off
Confusion matrix - VGG16	Analyse false-negative vs false-positive trade-off
Classification report - both models	Per-class precision / recall / F1 for each model
Model-comparison table & grouped bar chart	Side-by-side Custom CNN vs VGG16 across all metrics
Grad-CAM heatmaps	Explainability: verify decisions focus on lung fields
(Optional) ROC curves, error analysis	Extra rigour and qualitative error analysis

References

3. Kermany, D. S., et al. (2018). Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning. *Cell*, 172(5), 1122-1131.
4. Simonyan, K., & Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition (VGG). ICLR.
5. Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. ICCV.